

LIFE CYCLE ASSESSMENT

ISO 14044:2006 & ISO 14067:2018 Compliant



UNROOTED Mighty Ginger, Chilli & Baobab Shot 60ml

Happy Curations Ltd

Mighty Ginger blends ginger, turmeric & scotch bonnet chilli and baobab for a hit of spicy, caffeine-free natural energy and boost of immunity support. Thanks to baobab and ginger, each daily 60ml shot packs 20mg of vitamin C to give you natural good energy and fight tiredness and fatigue.

FUNCTIONAL UNIT

**1 unit of UNROOTED Mighty Ginger,
Chilli & Baobab Shot 60ml**

SYSTEM BOUNDARY

Cradle-to-Grave

REFERENCE YEAR

2025

01 Executive Summary

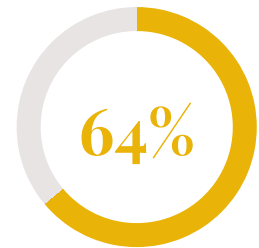
KEY INSIGHT

0.087

kg CO₂eq per unit

Fair data quality (64%)

DATA QUALITY



This Life Cycle Assessment evaluates the environmental impacts of UNROOTED Mighty Ginger, Chilli & Baobab Shot 60ml in accordance with ISO 14044:2006 and ISO 14067:2018. The assessment quantifies impacts across 6 environmental categories including climate change, water use, land use, acidification, eutrophication, and resource depletion. Data quality achieves a fair rating (64%) based on the ISO 14044 pedigree matrix approach, with 0 primary data points and 14 secondary database values.

CLIMATE IMPACT

0.087

kg CO₂e

WATER FOOTPRINT

0.025L

litres

CIRCULARITY

71%

recycling rate

02 Goal & Scope Definition

ISO 14044:2006 §4.2

Report Version 2.0

Intended Application

UNROOTED uses an LCA, conducted in accordance with ISO 14040/14044, to quantify environmental impacts across our full value chain — from baobab harvesting and raw material sourcing through to processing, packaging, and distribution. This methodology ensures comprehensive, standardised reporting across key impact categories including GWP (kg CO₂e), water, and land use, providing a credible, evidence-based foundation for our sustainability disclosures.

Reasons for Study

To quantify the carbon footprint and environmental impact of this product in accordance with ISO 14044/14067.

Intended Audience

Internal Stakeholders B2C Consumers
Investors & Shareholders General Public
B2B Customers

Comparative Assertion

No — this study does not support comparative assertions disclosed to the public.

System Boundary — Cradle-to-Grave

This assessment covers the complete product lifecycle from raw material extraction through manufacturing, distribution, consumer use, and final disposal or recycling, including end-of-life avoided burden credits for recycled packaging materials.

Cut-off Criteria

Mass: <1% of total input mass. Energy: <1% of total energy input. Environmental significance: Any flow contributing >1% to any impact category is included regardless of mass/energy contribution. Excluded: Vehicles and transport equipment; Buildings and infrastructure; Employee commuting; Office supplies and consumables; Business travel; Cleaning agents and lubricants; Maintenance materials; Laboratory chemicals; Stretch wrap and tertiary packaging

Allocation Procedure

Physical allocation by mass is applied for all co-products following the ISO 14044 Clause 4.3.4 hierarchy. Mass-based physical allocation was used throughout this assessment; no economic allocation was applied.

Assumptions & Limitations

- Assumption** System boundary: Cradle-to-Grave. Full lifecycle including raw materials, processing, packaging, distribution, consumer use phase, and end-of-life disposal/recycling.
- Assumption** Global warming potential (GWP-100) values from IPCC AR6: CO₂ = 1, CH₄ (fossil) = 29.8, CH₄ (biogenic) = 27.0, N₂O = 273 kg CO₂e/kg.
- Assumption** Functional unit: 1 unit of finished product as sold. All impacts are expressed per functional unit.

Emission factors sourced from a tiered hierarchy: (1) supplier specific verified data; (2) DEFRA 2025 conversion

02

Goal & Scope Definition (cont.)

Assumptions & Limitations (continued)

- Assumption
- Data quality scored using an impact-weighted average of material confidence scores. Materials without quality scores default to 40% ("Poor") per ISO 14044 §4.2.3.6.
- Assumption
- Facility overhead emissions (Scope 1 & 2) allocated to the product using mass-based allocation (production volume ratio). 1 production facility included.
- Assumption
- Electricity grid emission factors are country-specific (IEA 2023 / DEFRA 2025). Where the facility country is unknown, the global average (0.490 kg CO₂e/kWh) is used as a conservative fallback.
- Assumption
- Distribution transport emission factors sourced from DEFRA 2025: road freight ~0.104 kg CO₂e/tonne-km (HGV, average laden), rail ~0.028, container ship ~0.016, air freight ~1.130. Factors assume average load utilisation.
- Assumption
- Raw material transport impacts are embedded in the upstream emission factors from the database (cradle-to-gate factors include extraction, processing, and delivery to factory gate). Where ingredient-specific transport data is provided (origin, mode, distance), the generic embedded transport is replaced with DEFRA 2025 freight factors for the actual route. Electricity impacts are adjusted when the ingredient's processing country differs from the emission factor's electricity geography.
- Assumption
- Where supplier-specific data is unavailable, generic emission factors are used as proxies. This may over- or under-estimate impacts for specific supply chains.
- Assumption
- Capital goods (manufacturing equipment, buildings) and employee activities (commuting, business travel) are excluded per the stated cut-off criteria.

End-of-Life Modelling Assumptions (ISO 14044 §4.4)

Region: United Kingdom Method: Ecoinvent 3.12 recycling credits (avoided burden) Data Year: 2024

Material	Recycling	Landfill	Incineration	Composting	AD
Glass 60ml Bottle	74%	8%	18%	0%	0%
Label 60ml	74%	5%	15%	4%	2%
Aluminum Cap	72%	8%	20%	0%	0%
SRP packaging 60ml	28%	22%	50%	0%	0%
OUTER case cardboard	74%	5%	15%	4%	2%

Sources: DEFRA 2024, Ecoinvent 3.12, EU Packaging Directive, EPA 2024

03 Methodology

ISO 14044:2006 §4.3 & §4.4

Included Stages

- ✓ Raw material extraction & agricultural production
- ✓ Primary ingredient processing
- ✓ Packaging material manufacture
- ✓ Factory operations & energy use
- ✓ Distribution & transport to retail
- ✓ Consumer use phase (refrigeration)
- ✓ End-of-life disposal & recycling credits

Excluded Stages

- ✗ Capital goods & infrastructure
- ✗ Employee commuting

LCIA Method: ReCiPe 2016 v1.1 Midpoint (H)

ReCiPe 2016 v1.1 provides characterisation factors at the midpoint level using the Hierarchist cultural perspective. This method covers 18 impact categories with global scope. For climate change, the updated IPCC AR6 GWP-100 values (2021) are used in place of the original ReCiPe 2016 climate factors, consistent with current best practice (CO₂ = 1, CH₄ fossil = 29.8, CH₄ biogenic = 27.0, N₂O = 273). All other impact categories use the standard ReCiPe 2016 v1.1 characterisation factors.

Data Sources & Databases

Primary Supplier Data

Verified site-specific data from direct suppliers

0 factors

ecoinvent v3.12

Comprehensive LCI database for background processes

14 factors

AGRIBALYSE v3.2

ADEME agricultural and food product LCI database

7 factors

DEFRA Emission Factors v2025

UK government conversion factors for GHG reporting

5 factors

Software & Tools

AlkaTera Platform v2.0

LCA modelling, data collection, and report generation

ecoinvent v3.12

Background LCI data for industrial processes and materials

AGRIBALYSE v3.2

Agricultural and food product lifecycle inventory data (ADEME)

DEFRA Emission Factors v2025

UK government GHG conversion factors

03 Methodology — Characterization Models (continued)

LCIA CHARACTERIZATION FACTORS

IMPACT CATEGORY	CHARACTERIZATION MODEL	REFERENCE
Climate Change	IPCC AR6 GWP-100	IPCC Sixth Assessment Report (2021)
Acidification	Accumulated Exceedance	Seppälä et al. (2006), Posch et al. (2008)
Eutrophication (Freshwater)	EUTREND model	Struijs et al. (2009)
Eutrophication (Marine)	EUTREND model	Struijs et al. (2009)
Water Use	AWARE v1.3	Boulay et al. (2018)
Land Use	Soil Quality Index	Brandão & Milà i Canals (2013)
Resource Depletion	ADP ultimate reserves	van Oers et al. (2002)

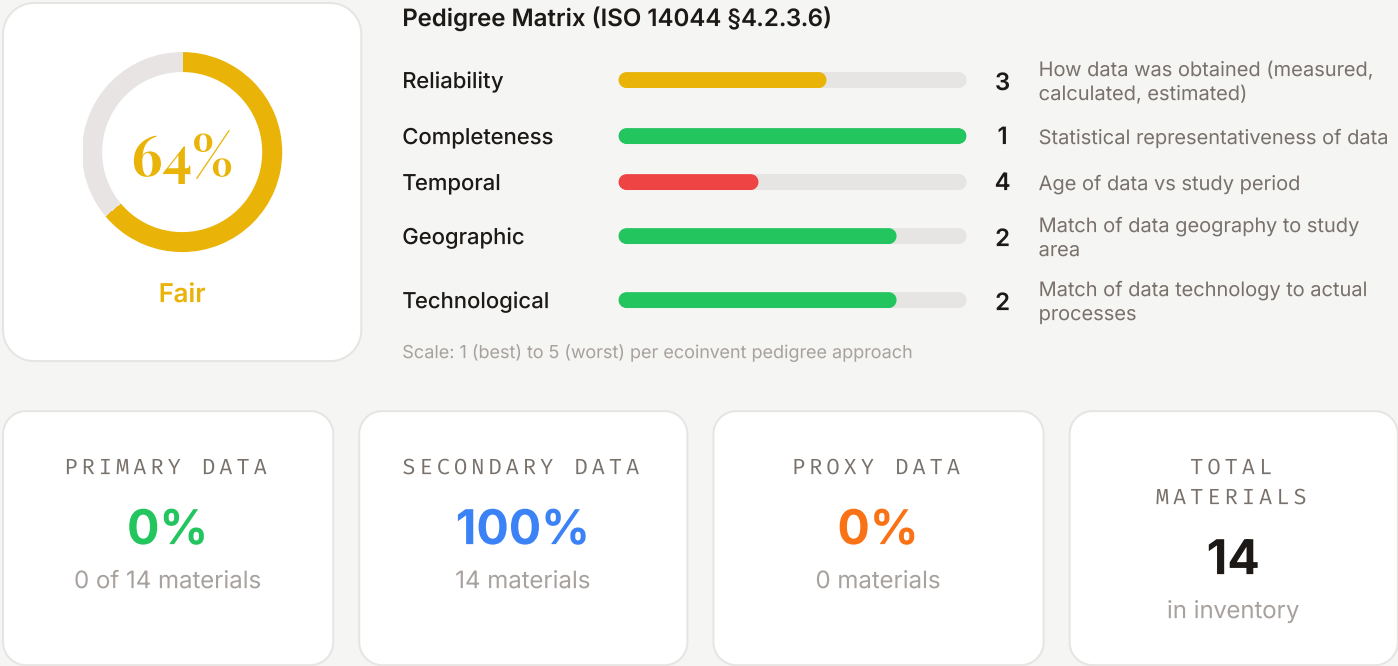
Reference Standards

- ISO 14040:2006 — Environmental management — Life cycle assessment — Principles and framework
- ISO 14044:2006 — Environmental management — Life cycle assessment — Requirements and guidelines
- ISO 14067:2018 — Greenhouse gases — Carbon footprint of products
- ISO 14046:2014 — Environmental management — Water footprint
- GHG Protocol — Product Life Cycle Accounting and Reporting Standard
- PAS 2050:2011 — Specification for the assessment of life cycle GHG emissions

04

Data Quality Assessment

ISO 14044:2006 §4.2.3.6 – DATA QUALITY REQUIREMENTS



Missing Data Treatment (ISO 14044 §4.2.3.6.3)

Missing data is handled using proxy emission factors from ecoinvent 3.12 and AGRIBALYSE 3.2 databases, selected based on geographic and technological representativeness. All proxy selections are documented with confidence scores and quality grades per ISO 14044 Clause 4.2.3.6.3.

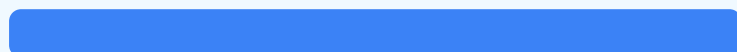
Uncertainty Assessment

Uncertainty in this assessment arises from: (1) measurement variability in primary data, (2) representativeness of secondary database values for the specific production system, and (3) characterisation model uncertainty in LCIA methods. A sensitivity analysis of key parameters is recommended for future iterations.

Data Quality Improvement Roadmap

- Priority 1** Collect primary data from top 3 impact contributors — this typically addresses >60% of total climate impact uncertainty.
- Priority 2** Replace proxy emission factors with material-specific secondary data from ecoinvent or AGRIBALYSE for improved geographic and technological representativeness.
- Priority 3** Verify transport distances and modes with suppliers to refine distribution stage estimates currently based on distance × freight factor calculations.
- Ongoing** Annual data refresh with updated DEFRA emission factors and database versions. Re-assess data quality scores after each improvement cycle.

Data Source Distribution



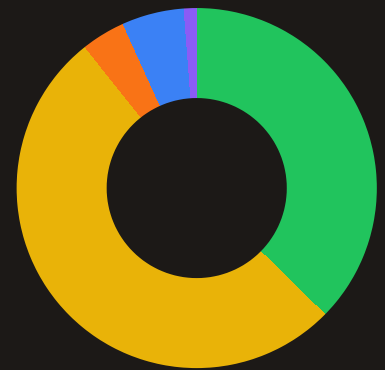
■ Primary 0% ■ Secondary 100% ■ Proxy 0%

05 Climate Impact

TOTAL CLIMATE IMPACT (GWP-100)

0.087 kg CO₂e

per 1 unit of UNROOTED Mighty Ginger, Chilli & Baobab Shot 60ml



LIFECYCLE STAGE BREAKDOWN

Raw Materials		0.0370 kg CO ₂ e	42.8%
Packaging		0.0515 kg CO ₂ e	59.5%
Distribution		0.0039 kg CO ₂ e	4.5%
Processing		0.0055 kg CO ₂ e	6.4%
Use Phase		0.0012 kg CO ₂ e	1.4%
End of Life		-0.0126 kg CO ₂ e	-14.6%

Raw Materials Packaging Distribution Processing Use Phase

Scope 1 (Direct)

0.0%

Scope 2 (Energy)

0.0%

Scope 3 (Value Chain)

100.0%

SCOPE ATTRIBUTION NOTE **CORRECT**

This product is manufactured at a third-party (contract) manufacturer (Natural Food and Drink Ltd, Unit 30 Britten Rd). Under GHG Protocol Product Standard §6.3.3 and Corporate Value Chain (Scope 3) Standard Category 1 (Purchased Goods and Services), the contract manufacturer's direct facility emissions (Scope 1) and purchased energy (Scope 2) are reported as Scope 3 Category 1 emissions of the commissioning brand. The zero values shown for Scope 1 and Scope 2 on this report are therefore correct: all production-stage emissions are captured within the Scope 3 / value-chain accounting, not omitted. This is a matter of attribution, not completeness.

05c

Processing & Manufacturing

PROCESSING EMISSIONS

0.0055

kg CO₂e per functional unit

SHARE OF PRODUCT FOOTPRINT

6.4%

of total lifecycle emissions

MANUFACTURING FACILITIES

1

1 contract

Natural Food and Drink Ltd, Unit 30 Britten Rd
(GB)

Contract Manufacturer (Scope 3)

Direct production run data

SCOPE 3 TOTAL

0.005098

kg CO₂e/unit

COMBUSTION

0.000000

kg CO₂e/unit

ELECTRICITY

0.005098

kg CO₂e/unit

All emissions from this contract manufacturer are classified as **Scope 3 Category 1** (Purchased Goods and Services) in the product footprint. The combustion/electricity split shows the emission source at the facility level.

ENERGY SOURCE	QUANTITY	UNIT	EMISSIONS (KG CO ₂ E)	SOURCE
Electricity (grid)	20250.00	kWh	4191.7500	Electricity

Attribution: 100.0% of facility output

Production volume: 822,204 units

Grid factor: 0.2070 kg CO₂e/kWh

Electricity: 20250.0 kWh (attributed)

Methodology

Processing emissions are allocated to the product using physical allocation by production volume (ISO 14044 Clause 4.3.4). Scope 1 factors from DEFRA 2025 GHG Conversion Factors; Scope 2 electricity from IEA/DEFRA 2023 country-specific grid emission factors.

Contract manufacturer emissions are classified as Scope 3 Category 1 (Purchased Goods and Services) per GHG Protocol Product Standard §6.3.3. Owned facility emissions are classified as Scope 1 (direct combustion) and Scope 2 (purchased electricity/heat).

06 Detailed GHG Reporting

ISO 14067:2018 — GREENHOUSE GAS BREAKDOWN

FOSSIL CO₂

0.0737

kg CO₂e

BIOGENIC CO₂

0.0129

kg CO₂e

LULUC CO₂

0.0000

kg CO₂e

GHG SPECIES	MASS (KG)	CO ₂ E (KG)	GWP-100
CO ₂ (fossil)	0.0737	0.0737	1
CO ₂ (biogenic)	0.0129	0.0129	1*
CO ₂ (LULUC)	0.0000	0.0000	1
CH ₄ (fossil)	0.000e+0	0.0000	29.8
CH ₄ (biogenic)	0.000e+0	0.0000	27.0
N ₂ O	0.000e+0	0.0000	273
HFCs / PFCs	—	0.0000	Variable
Total GWP-100 (all species)		0.0866 kg CO ₂ e	
Fossil Carbon Footprint (excl. biogenic)		0.0737 kg CO ₂ e	

Biogenic Carbon Note (ISO 14067:2018)

Biogenic CO₂ emissions are reported separately from fossil CO₂ per ISO 14067:2018 §6.4.9.3 and are excluded from the headline fossil carbon footprint. The value shown represents biogenic carbon released during processing, use, and end-of-life of biomass-derived inputs, as captured by the underlying emission factors (ecoinvent 3.12, AGRIBALYSE 3.2). Biogenic carbon uptake during biomass growth and any downstream release are accounted for inside those factors; this report does not apply an additional uptake credit or end-of-life virtual emission on top. Where upstream datasets already net uptake against release, the residual biogenic CO₂ shown here reflects that net. Land-use change (LULUC) carbon is reported on a separate line per ISO 14067 §6.4.9.4.

GWP Method: IPCC AR6 GWP100 — All GWP-100 values from IPCC Sixth Assessment Report (AR6, 2021)

* Per ISO 14067:2018 §6.4.9.3, biogenic CO₂ is characterised at GWP=1 for the species inventory but reported separately from the fossil carbon footprint. The net biogenic carbon balance (uptake minus end-of-life release) is excluded from the headline fossil footprint figure.

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Environmental Impact Categories

ISO 14044:2006 §4.4 – LIFE CYCLE IMPACT ASSESSMENT

Method: ReCiPe 2016 v1.1 Midpoint (H) — Impact category results are presented at the midpoint (characterisation) level without normalisation or weighting, in compliance with ISO 14044 Clause 4.4.3.4 requirements for public reporting.

Acidification
Terrestrial Acidification Potential

5.256e-4
kg SO₂-eq

Measures the potential to increase acidity of water and soil systems, primarily from SO₂, NO_x, and NH₃ emissions affecting ecosystems and infrastructure.

TOP CONTRIBUTORS

Glass 60ml Bottle	55.0%
Pineapple Juice NFC	25.8%
Ginger Juice NFC	19.2%

Eutrophication (Freshwater)
Freshwater Eutrophication Potential

2.256e-5
kg P-eq

Reflects excess nutrient enrichment (phosphorus) in freshwater systems, leading to algal blooms, oxygen depletion, and loss of aquatic biodiversity.

TOP CONTRIBUTORS

Glass 60ml Bottle	60.0%
Pineapple Juice NFC	27.8%
Aluminum Cap	12.2%

Fossil Resource Scarcity
Fossil Resource Depletion

2.703e-2
kg oil-eq

Quantifies the depletion of fossil fuel resources (oil, gas, coal) consumed across the product lifecycle.

TOP CONTRIBUTORS

Glass 60ml Bottle	68.1%
Pineapple Juice NFC	18.6%
Aluminum Cap	13.3%

08 Ingredient Impact Breakdown

PER-INGREDIENT ENVIRONMENTAL CONTRIBUTION • REAL INGREDIENT & CALCULATION FACTOR

INGREDIENT / CALC. FACTOR	QTY	ORIGIN	GWP (KG CO ₂ E)	% CLIMATE	ACID. (SO ₂ - EQ)	EUTROPH. (P-EQ)	DATA / CONF.
Scotch Bonnet Extract AGRIBALYSE 3.2	0.00020 kg	Ross-on- Wye HR9, UK	0.0002	0.2%	3.228e- 6	7.297e-8	Secondary 85%
Ginger Juice NFC AGRIBALYSE 3.2	0.01525 kg	Lima, Peru	0.0057	6.6%	7.340e- 5	8.973e-7	Secondary 85%
Pineapple Juice NFC AGRIBALYSE 3.2	0.01220 kg	Stuttgart, Germany	0.0153	17.7%	9.830e- 5	4.996e-6	Secondary 85%
Baobab Powder ecoinvent 3.12	0.00102 kg	Harare, Zimbabwe	0.0003	0.4%	1.950e- 6	9.300e-8	Secondary 85%
Lemon Juice NFC AGRIBALYSE 3.2	0.01017 kg	Murcia, Spain	0.0090	10.4%	6.486e- 5	1.874e-6	Secondary 85%
Water ecoinvent 3.12	0.02159 kg	Reading, UK	0.0000	0.0%	1.404e- 8	4.319e-9	Secondary 50%
Ascorbic Acid AGRIBALYSE 3.2	0.00002 kg	Shanghai, China	0.0000	0.0%	1.022e- 7	4.332e-9	Secondary 85%
Black Pepper Extract AGRIBALYSE 3.2	0.00001 kg	Mumbai Central, Mumbai, Maharashtra, India	0.0001	0.1%	3.431e- 7	1.264e-8	Secondary 85%
Glass 60ml Bottle ecoinvent 3.12	0.06000 kg	Rotherham, UK	0.0322	37.2%	2.100e- 4	1.080e-5	Secondary 50%
Label 60ml ecoinvent 3.12	0.00100 kg	Hungerford RG17, UK	0.0011	1.3%	6.500e- 6	3.500e-7	Secondary 50%

08

Ingredient Impact Breakdown

(continued)

INGREDIENT / CALC. FACTOR	QTY	ORIGIN	GWP (KG CO ₂ E)	% CLIMATE	ACID. (SO ₂ - EQ)	EUTROPH. (P-EQ)	DATA / CONF.
Aluminum Cap ecoinvent 3.12	0.00100 kg	Cranleigh GU6, UK	0.0092	10.6%	4.200e- 5	2.200e-6	Secondary 50%
SRP packaging 60ml ecoinvent 3.12	0.04800 kg	Hailsham, UK	0.0038	4.4%	2.200e- 5	1.120e-6	Secondary 50%
OUTER case cardboard ecoinvent 3.12	0.04700 kg	Hailsham, UK	0.0004	0.4%	2.154e- 6	1.097e-7	Secondary 50%
Turmeric Extract AGRIBALYSE 3.2	0.00002 kg	Mumbai, Maharashtra, India	0.0002	0.2%	6.986e- 7	2.757e-8	Secondary 85%

09 Water Footprint

TOTAL CONSUMPTION

0.025L
litres

SCARCITY-WEIGHTED

0.003L eq.
litres eq.

Water Sources

SOURCE	LOCATION	VOLUME	RISK LEVEL
Water	Reading, UK	0.022 L	LOW
Ginger Juice NFC	Lima, Peru	0.001 L	LOW
SRP packaging 60ml	Hailsham, UK	0.000 L	LOW
Pineapple Juice NFC	70 Stuttgart, Germany	0.000 L	LOW
Baobab Powder	Harare, Zimbabwe	0.000 L	LOW
Glass 60ml Bottle	Rotherham, UK	0.000 L	LOW
Lemon Juice NFC	Murcia, Spain	0.000 L	HIGH
Scotch Bonnet Extract	Ross-on-Wye HR9, UK	0.000 L	LOW
Label 60ml	Hungerford RG17, UK	0.000 L	LOW
Turmeric Extract	Mumbai, Maharashtra, India	0.000 L	HIGH
OUTER case cardboard	Hailsham, UK	0.000 L	LOW
Aluminum Cap	Cranleigh GU6, UK	0.000 L	LOW
Black Pepper Extract	Mumbai Central, Mumbai, Maharashtra, India	0.000 L	HIGH
Ascorbic Acid	Shanghai, China	0.000 L	MEDIUM
Processing & Facility Overheads	Allocated from production sites	0.002 L	LOW

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Circularity & Waste



Per-Material Disposal Pathways

MATERIAL	MASS	RECYCLE	LANDFILL	INCIN.	COMPOST	AD	NET (KG CO ₂ E)
Glass 60ml Bottle	0.060 kg	74%	8%	18%	0%	0%	-0.0149
Label 60ml	0.001 kg	74%	5%	15%	4%	2%	0.0002
Aluminum Cap	0.001 kg	72%	8%	20%	0%	0%	-0.0011
SRP packaging 60ml	0.004 kg	28%	22%	50%	0%	0%	0.0031
OUTER case cardboard	0.000 kg	74%	5%	15%	4%	2%	0.0001

Note: Recycled Content measures circular input (recycled material used in production). EoL Recycling Rate measures circular output (packaging recycled at end-of-life, based on regional defaults). These are independent metrics per ISO 14044 §4.4.5. Adapted from: Ellen MacArthur Foundation, Material Circularity Indicator (2015)

11 Land Use

TOTAL LAND USE

0.029m²
m² per year

Material Breakdown

MATERIAL	ORIGIN	MASS	INTENSITY	FOOTPRINT
Lemon Juice NFC	Murcia, Spain	0.010 kg	1.08	0.011 m ²
Pineapple Juice NFC	70 Stuttgart, Germany	0.012 kg	0.60	0.007 m ²
Ginger Juice NFC	Lima, Peru	0.015 kg	0.19	0.003 m ²
SRP packaging 60ml	Hailsham, UK	0.048 kg	0.05	0.002 m ²
Label 60ml	Hungerford RG17, UK	0.001 kg	0.90	0.001 m ²
Glass 60ml Bottle	Rotherham, UK	0.060 kg	0.01	0.001 m ²
Scotch Bonnet Extract	Ross-on-Wye HR9, UK	0.000 kg	2.54	0.001 m ²
Baobab Powder	Harare, Zimbabwe	0.001 kg	0.49	0.001 m ²
OUTER case cardboard	Hailsham, UK	0.047 kg	0.01	0.000 m ²
Turmeric Extract	Mumbai, Maharashtra, India	0.000 kg	7.76	0.000 m ²
Black Pepper Extract	Mumbai Central, Mumbai, Maharashtra, India	0.000 kg	7.71	0.000 m ²
Aluminum Cap	Cranleigh GU6, UK	0.001 kg	0.05	0.000 m ²

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Supply Chain

TOTAL DISTANCE

45102km

VERIFIED SUPPLIERS

0%

MATERIAL SUPPLIERS

Ginger Juice NFC Lima, Peru Transport: Sea freight	10113 km 0.01661 kg CO ₂ e
Ascorbic Acid Shanghai, China Transport: Sea freight	9244 km 0.00002 kg CO ₂ e
Baobab Powder Harare, Zimbabwe Transport: Sea freight	8307 km 0.00091 kg CO ₂ e
Turmeric Extract Mumbai, Maharashtra, India Transport: Sea freight	7257 km 0.00002 kg CO ₂ e
Black Pepper Extract Mumbai Central, Mumbai, Maharashtra, India Transport: Sea freight	7255 km 0.00001 kg CO ₂ e
Lemon Juice NFC Murcia, Spain Transport: Road (HGV)	1496 km 0.00164 kg CO ₂ e
Pineapple Juice NFC 70 Stuttgart, Germany Transport: Road (HGV)	782 km 0.00103 kg CO ₂ e
Glass 60ml Bottle Rotherham, UK Transport: Road (HGV)	223 km 0.00144 kg CO ₂ e
Scotch Bonnet Extract Ross-on-Wye HR9, UK Transport: Road (HGV)	123 km 0.00000 kg CO ₂ e
SRP packaging 60ml Hailsham, UK Transport: Road (HGV)	107 km 0.00055 kg CO ₂ e
OUTER case cardboard Hailsham, UK Transport: Road (HGV)	107 km 0.00054 kg CO ₂ e

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Interpretation (ISO 14044 §4.5)

Significant Issues

The most significant contributors to the carbon footprint are Glass 60ml Bottle, Pineapple Juice NFC, Lemon Juice NFC, collectively accounting for 68.0% of the total impact. The packaging stage dominates at 59.5%.

MATERIAL	IMPACT (KG CO ₂ E)	CONTRIBUTION
Glass 60ml Bottle	0.0330	38.1%
Pineapple Juice NFC	0.0159	18.4%
Lemon Juice NFC	0.0099	11.5%
Aluminum Cap	0.0092	10.6%
Ginger Juice NFC	0.0074	8.5%

Recycling Credits (Avoided Burden): End-of-life recycling generates 0.0167 kg CO₂e in avoided emissions by displacing virgin material production. Gross disposal emissions are 0.0041 kg CO₂e, giving a net EoL impact of -0.0126 kg CO₂e.

DOMINANT STAGE

Packaging
59.5% of total

DOMINANT SCOPE

Scope 3 (Value Chain)
100% of total

Key Findings

- The total carbon footprint is 0.087 kg CO₂e per functional unit.
- Glass 60ml Bottle is the largest single contributor at 38.1% of the total.
- The packaging stage accounts for over half the total impact (59.5%).
- Scope 3 (Value Chain) represents 100% of emissions, indicating the value chain is the primary source of impact.

Limitations

- Data quality compliance gaps identified: 1 issue(s).
- Secondary LCI data (ecoinvent, AGRIBALYSE, DEFRA) represents average technology and conditions; site-specific primary data would improve accuracy.

Recommendations

- Prioritise supplier engagement for Glass 60ml Bottle to obtain primary verified data and reduce uncertainty.
- Improve data quality for highly sensitive parameters (Glass 60ml Bottle) to strengthen result robustness.
- Conduct a third-party critical review per ISO 14044 §6 before using this study for public comparative assertions.

14

Uncertainty & Sensitivity (ISO 14044 §4.5.3)

PROPAGATED UNCERTAINTY

±22%

95% confidence interval

RESULT RANGE (95% CI)

0.0698 – 0.1074

kg CO₂e per functional unit

Sensitivity Analysis

±20% emission factor variation on top 3 contributors

MATERIAL	CONTRIBUTION	±20% RANGE	SENSITIVITY
Glass 60ml Bottle	38.1%	0.0800 – 0.0932 kg CO ₂ e	0.381 HIGH
Pineapple Juice NFC	18.4%	0.0834 – 0.0897 kg CO ₂ e	0.184
Lemon Juice NFC	11.5%	0.0846 – 0.0885 kg CO ₂ e	0.115

Conclusion: The total carbon footprint is most sensitive to Glass 60ml Bottle. A ±20% variation in this emission factor changes the total result by up to ±8%. Priority should be given to improving data quality for these materials.

Methodology: Uncertainty propagation follows the root-sum-of-squares approach for geometric standard deviation per Frischknecht et al. (2007), using the Pedigree Matrix (Weidema & Wesnæs, 1996) for data quality scoring. Sensitivity analysis applies ±20% variation to emission factors of the top three contributors.

15 Critical Review & Compliance Notes

Critical Review Status: NOT CONDUCTED

This LCA study has not undergone an independent critical review per ISO 14044 §6. A critical review by an external expert or panel is recommended before this study is used for public comparative assertions (ISO 14044 §5.3). The alkatera platform provides automated compliance checks but these do not constitute a formal critical review.

Recommendation: Engage a qualified independent reviewer with expertise in beverage product LCA to conduct a critical review per ISO 14044 §6.2 (external expert review) before publication.

Zero-Impact Categories

The following environmental impact categories were assessed but report zero values. Justification per ISO 14044 §4.4.2.2:

CATEGORY	REASON
Marine Eutrophication	No marine eutrophication characterisation factors available in the applied emission factor datasets for these input materials.
Ozone Depletion	No ozone-depleting substances identified in the product system.
Photochemical Ozone Formation	No photochemical ozone formation characterisation factors available for these input materials.
Particulate Matter	No particulate matter characterisation factors available for these input materials.
Human Toxicity (Cancer)	No carcinogenic toxicity characterisation factors available for these input materials.
Human Toxicity (Non-cancer)	No non-carcinogenic toxicity characterisation factors available for these input materials.
Primary Data Coverage	All 14 material/process inputs rely on secondary or proxy emission factors from databases (ecoinvent, AGRIBALYSE, DEFRA). No primary site-specific data has been collected. ISO 14044 §4.2.3.6 recommends increasing primary data coverage to improve representativeness. Collecting measured data from key suppliers (especially for the top 3 contributors) would significantly improve data quality.

Methodological Notes

Land use and land use change (LULUC) emissions are reported as zero. The current emission factor databases (ecoinvent 3.12, AGRIBALYSE 3.2, DEFRA) do not provide separate LULUC characterisation factors for all input materials. Where country-specific LULUC data becomes available, it will be incorporated in future assessments.

Inbound transport (supplier to factory) is calculated using DEFRA freight emission factors and added to each material's climate contribution during aggregation, allocated to the material's lifecycle stage per ISO 14044. Where OpenLCA decomposition replaced the factor's embedded generic transport with user-specific transport, it is counted once via the material's impact_climate instead. Outbound distribution (factory to retail/consumer) is only included when the system boundary extends to Cradle-to-Shelf or beyond.

Scope 1 emissions are allocated from facilities under operational control using physical allocation by production volume. Scope 2 emissions reflect purchased electricity and heat using the location-based method with country-specific IEA grid emission factors. Scope 3 emissions include upstream raw material extraction, processing, inbound transport, and contract manufacturing. Contract manufacturer emissions are classified as Scope 3 Category 1 (Purchased Goods and Services) per GHG Protocol §6.3.3.

Circularity Score Disclaimer: Weight-averaged recycled content across packaging materials. This is a proprietary simplification of the Ellen MacArthur Foundation Material Circularity Indicator (MCI v1.0) using only input-side data (recycled content %). It is not a certified MCI score.

15b AI-Assisted Internal Review

OVERALL RATING

NEEDS REMEDIATION

The study of UNROOTED Mighty Ginger, Chilli & Baobab Shot 60ml captures the main lifecycle stages but has one or more major gaps (most commonly: absent primary data or an incomplete inventory) that should be resolved before the report is used beyond internal exploratory analysis.

Findings by ISO 14044 / 14067 Clause

ISO 14044 §4.2 – Goal and scope definition

✓ CONFORMS

Functional unit, system boundaries, and reference flow are defined and consistent with the intended application.

Functional unit is "1 unit of UNROOTED Mighty Ginger, Chilli & Baobab Shot 60ml" with boundaries including 7 lifecycle stages. Cut-off criteria and allocation rules are disclosed in the methodology section.

ISO 14044 §4.2.3.6 – Data quality requirements

✗ MAJOR GAP

No primary verified data: all material inputs rely on secondary databases or proxy estimates.

Pedigree matrix assessment covers reliability, completeness, temporal, geographic, and technological representativeness. Current split: 0 primary / 14 secondary / 0 proxy across 14 inputs.

ISO 14044 §4.3.4 – Allocation procedures

✓ CONFORMS

Facility emissions are allocated to the product by physical allocation (production volume).

ISO 14044 requires step-wise allocation: avoidance by subdivision > system expansion > physical allocation > economic allocation. Physical allocation by production volume is applied where facility data is available.

ISO 14067:2018 §6.4.9.3 – Biogenic carbon accounting

✓ CONFORMS

Biogenic CO₂ is tracked separately from fossil CO₂ and reported as a memo item; biogenic CH₄ is reported with GWP-100 = 27.

No additional virtual end-of-life credit or debit is applied on top of background factors (ecoinvent / AGRIBALYSE already handle biogenic uptake and release within the factor).

GHG Protocol Product Standard §6.3.3 – Scope attribution

✓ CONFORMS

Zero Scope 1/2 is correct: product is contract-manufactured, so production-stage emissions are reported in Scope 3 Category 1 (Purchased Goods and Services).

ISO 14044 §4.2.3.4 – Completeness (inbound transport)

✓ CONFORMS

Inbound transport mode and distance are declared for all ingredients and are plausible for the stated origins.

ISO 14044 §4.5.3 – Sensitivity and uncertainty

! MINOR GAP

No formal Monte Carlo or scenario sensitivity analysis is included in this report.

For comparative assertions released to the public, ISO 14044 §5.3 requires uncertainty analysis. A Monte Carlo run over the top contributors (using pedigree-derived GSDs) is recommended before external use.



Our Commitment

This environmental assessment demonstrates our commitment to understanding and reducing the environmental impact of UNROOTED Mighty Ginger, Chilli & Baobab Shot 60ml. By measuring and reporting our footprint in accordance with ISO 14044 and ISO 14067 standards, we establish a transparent baseline for continuous improvement across our value chain.



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